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Linear equations

Keep equations balanced and solve for an unknown.

Mathematics 8-9 · algebra review

01 An equation is a balance

An equation states that two expressions have the same value. A variable represents an unknown number. A solution makes the original equation true when substituted.

02 Use inverse operations

Addition undoes subtraction. Division undoes multiplication. Apply the same operation to both sides; never divide by zero. For a two-step equation, usually undo addition or subtraction first.

03 Check the original equation

Substitute your answer into the original expression, not just your last line. Negative values and fractions can be valid solutions.

Worked example

Solve 3x + 4 = 19

Subtract 4 from both sides: 3x = 15. Divide both sides by 3: x = 5. Check: 3(5) + 4 = 19.

Watch out: 3x means 3 multiplied by x. Subtracting 3 does not undo multiplication by 3.

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Try it. Then check it.

Work through these questions before looking at the answers.

1. Solve x - 6 = 9.

2. Solve 2x + 5 = 17.

3. A group pays $6 plus $3 per notebook, for $24 in total. Find the notebook count.

Answer key

1. Add 6 to both sides: x = 15. Check: 15 - 6 = 9.

2. Subtract 5: 2x = 12. Divide by 2: x = 6.

3. Let n be the notebook count. 6 + 3n = 24, so 3n = 18 and n = 6 notebooks.

One-minute recap

Close the notes. Explain one key idea in your own words, then retry the question you found most difficult.

About this guide

Original Study Bridge topic review prepared with AI assistance. Suggested course fit, not complete or official course coverage. Compare with your teacher's materials. Examples and practice questions are newly written.

Official course context: Mathematics 8-9 · algebra review

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